A SEPARABLE ALGEBRAIC BASE CHANGE WITH SEPARABLY CLOSED RESIDUE FIELDS, IN WHICH SOLUTIONS DESCEND, EXISTS FOR ANY EXTENSION OF DISCRETE VALUATION RINGS

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Editorial note. The counted statement span includes the author's three-line commutative diagram, delivered verbatim. The three references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Solutions, weak solutions and separable solutions for an extension of discrete valuation rings. the property named in the statement at lines 34570-34572. Fixes what is being descended.
- Filtered colimits of finite etale extensions realizing a prescribed separable algebraic residue field extension. invoked at proof lines 34577-34579. Produces A' with the required residue field and the colimit structure on which the descent argument relies.
- Strict henselization and the fact that a finite etale algebra over a strictly henselian local ring is a product of copies of that ring. used at proof lines 34585-34598. Is what makes B' tensor_A A'_1 split into copies of B', which is the key step of the descent.
- The solutions-go-down lemma. invoked at proof lines 34604-34605. Carries the solution from A -> B' to A -> B.

Source reference. Original lemma 09E0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11883–11883. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 04GT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5844–5844. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 34517: sources/stacks-more-algebra/pinned-original-source.tex, source lines 34517–34517. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $A \to B$ be an extension of discrete valuation rings. There exists a commutative diagram*

$$\begin{array}{ccc} B & \longrightarrow & B' \\ \uparrow & & \uparrow \\ A & \longrightarrow & A' \end{array}$$

of extensions of discrete valuation rings such that

(1) *the extensions K'/K and L'/L of fraction fields are separable algebraic,*
(2) *the residue fields of A' and B' are separable algebraic closures of the residue fields of A and B, and*
(3) *if a solution, weak solution, or separable solution exists for $A' \to B'$, then a solution, weak solution, or separable solution exists for $A \to B$.*

Proof. By Algebra, Lemma 09E0 there exists an extension $A \subset A'$ which is a filtered colimit of finite étale extensions such that the residue field of A' is a separable algebraic closure of the residue field of A. Then $A \subset A'$ is an extension of discrete valuation rings such that the induced extension K'/K of fraction fields is separable algebraic.

Let $B \subset B'$ be a strict henselization of B. Then $B \subset B'$ is an extension of discrete valuation rings whose fraction field extension is separable algebraic. By Algebra, Lemma 04GT there exists a commutative diagram as in the statement of the lemma. Parts (1) and (2) of the lemma are clear.

Let K'_1/K' be a (weak) solution for $A' \to B'$. Since A' is a colimit, we can find a finite étale extension $A \subset A'_1$ and a finite extension K_1 of the fraction field F of A'_1 such that $K'_1 = K' \otimes_F K_1$. As $A \subset A'_1$ is finite étale and B' strictly henselian, it follows that $B' \otimes_A A'_1$ is a finite product of rings isomorphic to B'. Hence

$$L' \otimes_K K_1 = L' \otimes_K F \otimes_F K_1$$

is a finite product of rings isomorphic to $L' \otimes_{K'} K'_1$. Thus we see that K_1/K is a (weak) solution for $A \to B'$. Hence it is also a (weak) solution for $A \to B$ by Lemma 34517. □